HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Collected, validated, and analysed sustainability data including carbon, energy, water, and waste metrics for quarterly environmental performance reports.
- Supported the creation of sustainability performance dashboards, contributing evidence and tracking progress against internal targets for resource efficiency initiatives.
- Researched emerging sustainability legislation and best practices in the water sector, compiling findings into actionable guidance documents for project teams.
- Collaborated with project delivery and design teams to embed practical environmental considerations into early-stage project planning and execution.
- Developed Python scripts to automate the aggregation of disparate operational data sources, reducing manual data collation effort by approximately **15 hours** per month.
- Contributed to the development of a **Java Spring Boot** microservice for internal data validation, ensuring adherence to data quality standards for downstream reporting.
- Managed and maintained CI/CD pipelines using **Jenkins** for automated build, test, and deployment of internal tools, reducing average deployment time by **30%**.

SKILLS

Sustainability Concepts – carbon reduction, circular economy, biodiversity, social value, responsible procurement, environmental, social and governance (ESG)

Data Analysis & Reporting – data validation, performance dashboards, target tracking, data visualisation, reporting tools

Programming & Scripting – Python, Java, SQL

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Core Technologies – Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Jira, Confluence, PostgreSQL, MySQL, Oracle

Domain Knowledge – water and wastewater infrastructure, environmental management, energy management

PROJECTS

Environmental Data Aggregation and Reporting Platform

- Developed a **Python**-based data aggregation pipeline to consolidate environmental performance data from multiple operational systems into a central repository.
- Engineered **SQL** queries to extract and transform data for use in sustainability reporting, ensuring accuracy and consistency across different data sources.
- Designed and implemented automated data validation checks within the pipeline, catching over **90%** of data integrity issues before they reached reporting stages.

Internal Tooling for Resource Efficiency Tracking

- Built a **Java Spring Boot** application with a **REST API** to manage and track resource efficiency initiatives across different business units.
- Integrated the application with an existing **PostgreSQL** database for persistent storage of initiative data and performance metrics.
- Containerised the application using **Docker** for streamlined deployment and management on **AWS EC2** instances.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)